

No Bias Left Behind: Fairness Testing for Deep Recommender Systems Targeting General Disadvantaged Groups

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Recommender systems play an increasingly important role in modern society, powering digital platforms that suggest a wide array of content, from news and music to job listings, and influencing many aspects of daily life. To improve personalization, these systems often use demographic information. However, ensuring fairness in recommendation quality across demographic groups is challenging, especially since recommender systems are susceptible to the “rich get richer” Matthew effect due to user feedback loops. With the adoption of deep learning algorithms, uncovering fairness issues has become even more complex. Researchers have started to explore methods for identifying the most disadvantaged user groups using optimization algorithms. Despite this, suboptimal disadvantaged groups remain underexplored, which leaves the risk of bias amplification due to the Matthew effect unaddressed. In this paper, we argue for the necessity of identifying both the most disadvantaged and suboptimal disadvantaged groups. We introduce **FairAS**, an adaptive sampling based approach, to achieve this goal. Through evaluations on four deep recommender systems and six datasets, FairAS demonstrates an average improvement of 19.2% in identifying the most disadvantaged groups over the state-of-the-art fairness testing approach (FairRec), while reducing testing time by 43.07%. Additionally, the extra suboptimal disadvantaged groups identified by FairAS help improve system fairness, achieving an average improvement of 70.27% over FairRec across all subjects.

CCS Concepts: • **Software and its engineering** → **Search-based software engineering; Software testing and debugging**; • **Information systems** → **Recommender systems**.

Additional Key Words and Phrases: Fairness Testing, Deep Recommender System, Search-based Software Testing

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1 Introduction

Recommender systems (RSs) are designed to generate tailored, personalized recommendations for users, and they are increasingly popular as digital platforms grow, reaching vast, diverse user populations. These systems offer recommendations across a wide range of objects, from information

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(such as news), to products (like music), and even opportunities (like jobs). With so many options available, accurately personalizing recommendations is crucial to providing users with relevant results. To achieve this, RSs often use demographic information, including factors like gender, race, and age, to guide recommendations. However, ensuring that recommendations maintain high quality and fairness across different demographic groups is essential to prevent group-wise fairness issues. A lack of fairness can lead to disparities, where certain groups consistently receive lower-quality recommendations. This limitation will reinforce the systemic bias and contribute to the “rich get richer, and poor get poorer” phenomenon, also known as the Matthew effect, where initial inequalities in recommendations are reinforced over time.

Uncovering fairness issues in recommender systems—particularly those driven by black-box deep learning algorithms (also called deep recommender systems)—is challenging and costly, largely due to the vast size of the user base. Recently, researchers have begun to develop more efficient approaches to reveal and quantify these fairness problems. To our knowledge, FairRec [19] is the first approach specifically tailored to fairness testing in deep recommender systems. It focuses on identifying fairness issues across user groups, which are segmented based on sensitive attributes like gender, race, and age, with particular emphasis on measuring the maximum disparity in recommendation quality between *the most advantaged groups* and *the most disadvantaged groups*, i.e., groups with the best and worst recommendation quality. To achieve this, FairRec employs a global optimization technique, discrete particle swarm optimization (DPSO), to expedite the discovery of relevant groups.

However, the approach overlooks *suboptimal disadvantaged groups*—those whose recommendation quality falls below the system’s overall performance but not as low as the most disadvantaged group. Given the Matthew effect’s impact on recommender systems, the biases experienced by these suboptimal groups may be amplified over time. Specifically, RSs collect user feedback to improve their recommendations. However, determining whether a recommendation is genuinely good can be challenging. When users assess the relevance of a recommendation, they often engage with it similarly to how they engage with recommendations they favor. This presents a long-standing challenge for recommender systems. When recommendations are biased, users are frequently confronted with many low-quality options. For example, they may read through full job descriptions only to discover that they are not interested in the positions. For these groups, where the recommended content often does not align with their true preferences, the noise in the collected feedback hinders the system from automatically correcting its recommendations. As a result, the model continues to learn from these low-quality data within the feedback loop, leading to a persistent decline in the quality of its recommendations. Additionally, suboptimal disadvantaged groups, especially those with relatively small disparities from the most disadvantaged group, play unique roles when enhancing overall system fairness through retraining or post-processing techniques [11, 19, 50], as further supported by our evaluation (see Section 5.2). Hence, we argue that identifying both the most disadvantaged and suboptimal disadvantaged groups is essential, and there is a clear need for an effective and efficient approach to achieving this.

In this work, we introduce **FairAS** (Fairness testing via Adaptive Sampling), the first approach designed to achieve dual fairness testing objectives through adaptive sampling [36]. Traditional global optimization algorithms are typically focused on finding the optimal target, which can make them prone to overlooking diverse suboptimal solutions. This characteristic makes them less suitable for our goal of comprehensive fairness testing. Instead, we employ sampling algorithms, which provide more robust and efficient exploration of the testing space [29, 35], making them an ideal fit for our task. User groups are segmented by values of sensitive attributes (e.g., gender and age), and effectively sampling these attribute values is crucial for identifying disadvantaged groups. Specifically, FairAS maintains a selection probability for each value of each attribute. The

key insight of adaptive sampling is that as sampling progresses, the distribution of the sample space becomes more apparent, allowing for dynamic adjustments to the sampling strategy. In FairAS, once a group is sampled, we evaluate its recommendation quality. As more groups are assessed, the fairness distribution among them becomes clearer. By examining the patterns of disadvantaged groups, we gain insights into attribute values that are more likely to contribute to unfairness. Consequently, FairAS updates the sampling probability for each attribute value after each round of sampling. Additionally, we observed that disadvantaged groups tend to cluster, which inspired us to conduct a local search around each newly identified disadvantaged group in each sampling round. This local search explores groups with slightly different attribute values, effectively accelerating the testing process, as shown in our evaluation (Section 3.2.1). Notably, the sampling strategy provides multiple starting points for local search, facilitating simultaneous identification of both the most disadvantaged and suboptimal disadvantaged groups.

To evaluate the effectiveness of FairAS, we conducted an extensive study involving four deep recommender systems from Google and Huawei, using six diverse datasets. This setup yielded 24 evaluation subjects (four deep recommender systems $\times$ six datasets). First of all, our experimental results demonstrate that FairAS outperforms the state-of-the-art FairRec in both effectiveness and efficiency when identifying the most disadvantaged group. On average, FairAS improves effectiveness by 19.2% over FairRec while reducing running time by 43.07% across all subjects. Further, FairAS more accurately identifies suboptimal disadvantaged groups compared to FairRec, achieving effectiveness closer to the ground-truth results obtained by brute-force search, with an average detection rate of 81.98% for the top 10% of suboptimal disadvantaged groups. Then, FairAS substantially enhances system fairness based on these identified suboptimal groups, with an average improvement over FairRec of 70.27% across all subjects.

In summary, this paper makes the following main contributions:

- We extend the objective of existing fairness testing for deep recommender systems to include the identification of both the most disadvantaged and suboptimal disadvantaged groups.
- We propose the first approach, called FairAS, designed to achieve this important yet challenging objective through adaptive sampling enhanced by local search.
- We conducted an extensive study on 24 subjects, demonstrating FairAS's superior performance in identifying targeted disadvantaged groups and enhancing system fairness based on those identified groups.

Note that this work is largely relevant to software engineering (SE). Besides the relevant problem of testing, search-based methods (including the ideas of adaptive sampling and local search employed in FairAS) have been widely used in many other SE tasks [9, 18, 29, 43, 48]. Although our evaluation does not include SE-relevant datasets due to no such datasets released in this field, the methodology design of FairAS is general. In the future, once such datasets are available, we can apply FairAS to them for making more SE-relevant contributions.

2 Background

2.1 Deep Recommender Systems

A recommender system aims to provide the items that a user may be interested in based on the personal profile and historical behavior of the user. It consists of three main components: user encoder, item encoder, and interaction predictor. Specifically, it first extracts and encodes user preference features, which include both sensitive attributes (e.g., gender, age, or ethnicity) and insensitive attributes (e.g., click history). Then, it extracts and encodes item features, which can include various types of data (such as text and images). Finally, it predicts the interaction probability of each item for a user based on both user and item features and then ranks items for

recommendation. Different from traditional statistical/collaborative filtering-based recommender systems, deep recommender systems employ deep neural networks to encode both user and item features. As shown in the existing work [51], deep recommender systems have become more mainstream in practice. For example, Google proposed Wide&Deep [12] for application recommendations in Google Play, a mobile app store with over one billion active users and over one million applications.

2.2 Fairness of Recommender Systems

In this paper, we focus on group fairness of deep recommender systems. Here, a group refers to a set of users with the same setting in *sensitive attributes*, and different groups should be treated with little distinction in terms of an evaluation metric focused by a recommender system. Sensitive attributes refer to users' personal information that has the potential to subject them to discrimination or unfair treatment, examples of which include gender, age, or occupation [15].

Given an evaluation metric M , the measurement of the recommendation results received by a single group G is calculated as $M(G) = \frac{1}{|G|} \sum_{u \in G} M(u)$. The evaluation metrics used in our study will be introduced in Section 2.3. We call the groups that have lower scores compared to the system's overall performance in terms of the evaluation metric *disadvantaged groups*, which are more likely to experience negative outcomes or biases, and the opposite groups *advantaged groups*, which are more likely to benefit from favorable outcomes or privileges. Among all disadvantaged groups, there are both the most disadvantaged and suboptimal disadvantaged groups, with the group G having the minimum $M(G)$ value defined as *the most disadvantaged group*, while the others are considered *suboptimal disadvantaged groups*. The disadvantaged groups with lower scores in terms of the evaluation metric indicate *more severe* fairness issues.

Current research [19] focuses on identifying the most disadvantaged group, targeting the elimination of the most severe fairness issue. However, suboptimal disadvantaged groups are equally important, as they represent a broader spectrum of unfairness cases, uncovering additional discriminated groups that require attention. Addressing these suboptimal groups is often more impactful for improving the overall fairness of a recommender system than focusing solely on the most disadvantaged group. Furthermore, identifying suboptimal groups with lower scores can help address more severe fairness issues, further enhancing their critical significance. In this work, we extend the objective of existing fairness testing for deep recommender systems to include identifying both the most disadvantaged and suboptimal disadvantaged groups, given a specific evaluation metric and a limited testing budget. Deep recommender systems typically involve multiple sensitive attributes, each with a large set of possible values, resulting in an enormous number of potential groups. Consequently, achieving this extended objective within such a huge space is highly challenging.

2.3 Evaluation Metrics

There exist numerous metrics that assess the recommendation quality across various dimensions, including recommendation accuracy, diversity, and popularity. In our study, we evaluate the effectiveness of our proposed fairness testing approach under different metrics to ensure evaluation generalizability. Following the previous research [19], we consider the five popular metrics: AUC (Area under the Curve) [27], MRR (Mean Reciprocal Rank) [40], NDCG@ k (Normalized Discounted Cumulative Gain for the top k recommendations) [23], URD (User Recommendation Diversity) [33], and URP (User Recommendation Popularity) [4]. The first three metrics measure recommendation accuracy, while the latter two measure recommendation diversity and popularity respectively.

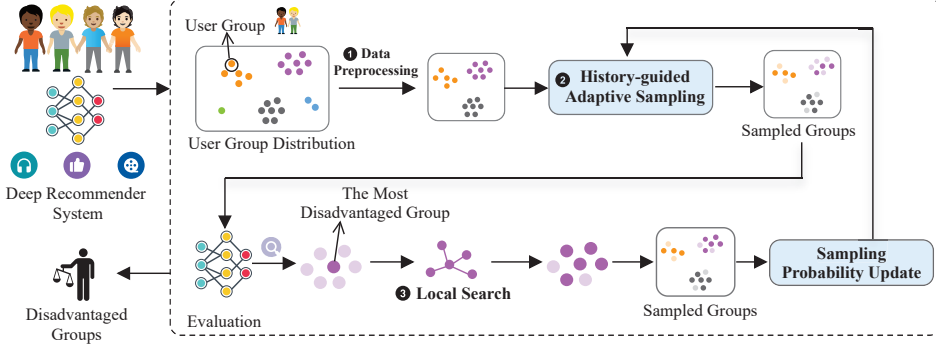


Fig. 1. Overview of FairAS

- **AUC** is determined by calculating the area beneath the Receiver Operating Characteristic (ROC) curve. This curve illustrates a model's performance by graphing the true positive rate versus the false positive rate at various thresholds.
- **MRR** is calculated using the formula: $MRR = \frac{1}{|U|} \sum_{i=1}^{|U|} \frac{1}{pos(i)}$, where $|U|$ denotes the total number of users, and $pos(i)$ represents the rank of the favorite item in the recommendation list for user u_i .
- **NDCG@k** is determined using the formula $NDCG@k = \frac{DCG@k}{IDCG@k}$. Here, $DCG@k$ is calculated as $DCG@k = \sum_{i=1}^k \frac{2^{p(i)} - 1}{\log_2(i+1)}$, where $p(i)$ represents the preference score of the user u for the i^{th} item in the generated recommendation list. $IDCG@k$ represents the DCG value of the ideal k -item recommendation list.
- **URD** measures the degree of recommendation diversity. It is calculated based on the intra-list similarity [53], using the formula: $URD(rl) = 1 - \frac{2}{n(n-1)} \sum_{i_a \in rl} \sum_{i_b \neq i_a \in rl} Sim(i_a, i_b)$, where rl denotes a recommendation list with n items, and $Sim(i_a, i_b)$ represents the similarity between item i_a and item i_b .
- **URP** measures the consistency between the popularity of the recommended items and the preference of the user. It is calculated as $URP(rl, hl) = \frac{1}{|rl|} \sum_{i_j \in rl} RP_j - \frac{1}{|hl|} \sum_{i_j \in hl} RP_j$, where rl refers to the recommendation list and hl refers to the items the user feels interest in. RP_j represents the popularity of the j^{th} item (i_j), which is calculated as $RP_j = \frac{C(i_j)}{|T|}$, where $C(i_j)$ denotes the number of interactions on i_j in the training set, and $|T|$ denotes the total number of interactions across all items in the training set.

3 Approach

We propose a novel fairness testing approach, called FairAS, to identifying both the most disadvantaged and suboptimal disadvantaged groups efficiently, based on adaptive sampling enhanced by local search. Figure 1 shows the overview of FairAS. Specifically, before performing fairness testing, FairAS first eliminates invalid groups, defined as those whose user counts are less than a threshold following the existing work [19], to largely reduce the testing space, since visiting such user groups could be a waste of time (Section 3.1). Then, FairAS designs an adaptive sampling strategy to explore the remaining testing space based on exploration history and local search, in order to accelerate the identification of both the most disadvantaged and suboptimal groups (Section 3.2). Finally, FairAS returns the identified disadvantaged groups when reaching the terminating condition (i.e., the most disadvantaged group identified do not change for five consecutive iterations).

3.1 Data Preprocessing

Given a deep recommender system, the fairness testing process requires to frequently visit user groups for evaluation. However, there are a large number of invalid groups in the space, causing that a substantial amount of time could be wasted on exploring them. Following the existing work [19], a user group is deemed invalid if the number of users within this group is smaller than a predefined threshold τ . To improve testing efficiency, we design a data preprocessing strategy to filter out invalid groups in advance.

Intuitively, before the sampling process, we can first construct a forbidden collection recording invalid groups, and then the corresponding space could be reduced. A natural solution to build this forbidden collection can be identifying invalid groups by enumerating all user groups. However, it can be extremely time-consuming since the number of user groups could grow exponentially with the number of sensitive attributes increasing. Hence, to efficiently reduce the space, instead of enumerating all user groups, we propose a lightweight filtering strategy that focuses on examining single-attribute and binary-attribute groups. This aligns with the idea of combinatorial interaction testing, where achieving high t -wise coverage is infeasible for large t values and $t \in \{1, 2\}$ is often adopted in practice [6].

Given a deep recommender system R , a set of users U , and a set of sensitive attributes S , a single-attribute group can be represented as follows:

$$G^1(s_i = x_m) = \{u \in U \mid \phi(u, s_i) = x_m\} \quad (1)$$

where $s_i \in S$, and $\phi(u, s_i)$ denotes the value of the attribute s_i for user u . We use $G^1(s_i = x_m)$ to denote the single-attribute group of all users in which the value of attribute s_i is x_m . Similarly, a binary-attribute group can be defined as follows:

$$\begin{aligned} G^2(s_i = x_m \wedge s_j = y_k) = \\ \{u \in U \mid \phi(u, s_i) = x_m \wedge \phi(u, s_j) = y_k\} \end{aligned} \quad (2)$$

Here, $G^2(s_i = x_m \wedge s_j = y_k)$ is actually one subset of $G^1(s_i = x_m)$, because condition $s_i = x_m$ subsumes condition $s_i = x_m \wedge s_j = y_k$.

If an invalid single-attribute group G^1 is identified, it can be confidently asserted that any multi-attribute group stemming from G^1 will also be invalid. This principle equally holds for binary-attribute groups. With this data preprocessing component, FairAS returns a reduced set of user groups G^n for the subsequent sampling process.

3.2 Local-Search-Enhanced Adaptive Sampling

3.2.1 Top-Level Design. As presented before, the global optimization process like FairRec struggles with identifying suboptimal disadvantaged groups and suffers from slow convergence. Instead, sampling algorithms can offer broader, more efficient exploration [29, 35]. Therefore, we design a sampling-based strategy in FairAS for achieving our goal of identifying both the most disadvantaged and suboptimal disadvantaged groups efficiently. Specifically, FairAS guides the sampling process by dynamically updating the sampling probability for setting each sensitive attribute based on exploration history (to be presented in Section 3.2.2). This is because exploration history can provide directional cues for potential disadvantaged groups in follow-up samplings.

To further improve efficiency within the large sampling space, FairAS incorporates local search into the adaptive sampling process (to be presented in Section 3.2.3), based on the observation that disadvantaged groups tend to cluster. This clustering allows local searches around an identified disadvantaged group to accelerate the detection of additional, potentially more severe, disadvantaged groups. The adaptive sampling strategy provides multiple starting points for local search, facilitating

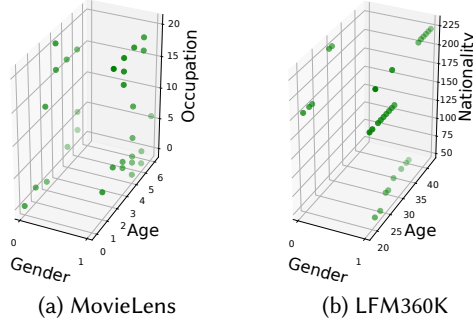


Fig. 2. Clustered distribution of disadvantaged groups in the testing space.

simultaneous identification of both the most disadvantaged and suboptimal disadvantaged groups. Specifically, after each sampling iteration, FairAS performs local search on the disadvantaged group with the lowest evaluation metric value among the sampled groups. These groups visited via local search are then added to the exploration history to update probabilities, guiding the next sampling iteration. In this way, sampling and local search are interwoven throughout the process.

We have conducted a preliminary study to investigate the feasibility of using local search for accelerating fairness testing in deep recommender systems. Specifically, we took the Wide&Deep model on two datasets: MovieLens and LFM360K (detailed in Section 4.1) as the representatives. We enumerated all valid user groups (setting the filtering threshold for invalid group τ to 0.1% of the user population following the existing work [19]), and then calculated the AUC value (described in Section 2.3) for each user group. We visualized the 30 groups with the smallest AUC values (representing disadvantaged groups) in Figure 2, where the x -axis, y -axis, and z -axis correspond to three sensitive attributes, respectively. From Figure 2, it becomes evident that the disadvantaged groups tend to cluster together in the space. This clustering behavior aligns with the principles of local search, suggesting its potential effectiveness in expediting the identification of the most disadvantaged and suboptimal disadvantaged groups.

3.2.2 History-guided Adaptive Sampling. FairAS iteratively samples user groups to find the most disadvantaged and suboptimal disadvantaged groups. During each iteration, FairAS samples k user groups for evaluation, each of which is required to set the value for each sensitive attribute through adaptive probabilistic sampling. Specifically, for each sensitive attribute, FairAS assigns a probability to each value in its value set, and the value with a larger probability is more likely to be set, indicating that it is more helpful to find more disadvantaged groups. Given a set of sensitive attributes $S = \{s_1, s_2, \dots, s_n\}$, we denote the set of values for s_i as $X_i = \{x_{i1}, x_{i2}, \dots, x_{im_i}\}$ and the probability that s_i 's value is sampled as x_{ij} as $\text{prob}(x_{ij})$.

Initially, FairAS assigns an equal probability for all the values corresponding to a sensitive attribute. In each of subsequent iterations, FairAS needs to update the probability of each value for each sensitive attribute according to the already explored valid groups before this iteration in order to improve efficiency. The updating strategy consists of two steps: (1) First, FairAS ranks all the previously explored valid groups $\mathbb{G}_{\text{explored}}$ in the descending order of their metric values (e.g., AUC values) produced by the deep recommender system under test (Line1-2). FairAS regards the latter half of groups in the ranking list as potential disadvantaged groups $\mathbb{G}_{\text{dis}}$ and the sensitive-attribute values occurring in them are more likely to help find the most disadvantaged and suboptimal disadvantaged groups (Line3-4). (2) For each sensitive attribute s_i ($1 \leq i \leq n$), FairAS counts the number of occurrences of each value x_{ij} ($1 \leq j \leq m_i$) in these potential disadvantaged groups,

Algorithm 1: Disadvantaged Group Sampling Probability Update

Input : $\mathbb{G}_{explored}$: the set of all valid groups explored in history
 S : the set of sensitive attributes
Output: $prob$: updated sampling probability for the next iteration

```

1  $M \leftarrow evaluation(\mathbb{G}_{explored})$ 
2  $G_{sorted} \leftarrow reverseSort(\mathbb{G}_{explored}, M)$ 
3  $num \leftarrow len(G_{sorted})$ 
4  $G_{dis} \leftarrow G_{sorted}[\lfloor num/2 \rfloor :]$ 
5 foreach  $s_i$  in  $S$  do
6   foreach  $x_{ij}$  in  $X_i$  do
7      $t_{ij} \leftarrow countFrequency(s_i, x_{ij}, G_{dis})$ 
8      $prob(x_{ij}) \leftarrow \frac{t_{ij}}{\sum_{j=1}^m t_{ij}}$ 
9   end
10 end
11 return  $prob$ 

```

Algorithm 2: Local Search

Input : $\mathbb{G}_{explored}$: the set of all valid groups explored in history
 $\mathbb{G}_{sampled}$: the set of valid groups sampled by history-guided sampling in the last iteration
 T : the number of exploration times for each attribute
Output: $\mathbb{G}_{explored}$: updated set of explored user groups after local search

```

1  $\mathbb{G} \leftarrow \emptyset$ 
2  $M \leftarrow evaluation(\mathbb{G}_{sampled})$ 
3  $G_{local} \leftarrow getMinGroup(\mathbb{G}_{sampled})$  /* get the group with the minimum metric value among the sampled groups
   in the last iteration */
4 foreach  $s_i$  in  $S$  do
5    $T \leftarrow min(T, |X_i| - 1)$ 
6    $X_{T_i} \leftarrow randomSample(X_i, T)$ 
7   foreach  $t$  in  $X_{T_i}$  do
8      $G' \leftarrow randomChange(G_{local}, s_i, t)$ 
9      $\mathbb{G} \leftarrow \mathbb{G} \cup \{G'\}$ 
10  end
11 end
12  $\mathbb{G}_{explored} \leftarrow \mathbb{G} \cup \mathbb{G}_{explored}$ 
13 return  $\mathbb{G}_{explored}$ 

```

denoted as t_{ij} (Line5-7). With the intuition that the more frequently x_{ij} occurs, the more relevant x_{ij} is to find the most disadvantaged and suboptimal disadvantaged groups, FairAS updates the probability of x_{ij} , $prob(x_{ij})$, as $\frac{t_{ij}}{\sum_{j=1}^m t_{ij}}$ (Line8). Finally, FairAS samples k user groups according to these updated probabilities in this iteration.

3.2.3 Local Search Enhancement. After obtaining a set of user groups through history-guided adaptive sampling in each iteration, denoted as $\mathbb{G}_{sampled}$, FairAS starts the local search process. We use $\mathbb{G}_{explored}$ to denote all valid groups sampled in history (including $\mathbb{G}_{sampled}$). Within this context, FairAS selects the most disadvantaged group among the set $\mathbb{G}_{sampled}$ as the target for local search, denoted as G_{local} (Line2-3). Specifically, FairAS systematically examines each sensitive attribute

Table 1. Statistics information of datasets

Dataset	#Items	#Interactions	#Users	#User Groups	Sensitive Attributes
LFM360K	140,000	17 million	267,034	53,058	gender, age, nationality
MovieLens	3,883	1 million	6,040	294	gender, age, occupation
BlackFriday	3,566	550,068	5,891	8,820	gender, age, occupation, city category, years of stay in the current city, marital status
Amazon	29,391	944,347	188,657	1,000	the number of purchases, total consumption, the highest price of items purchased
Reasoner	4,672	58,497	2,997	468,720	age, gender, education, career, income, address
Kuai	10,728	12,530,806	7,176	87,250,800	seven user personal attributes

* In Kuai, to protect user privacy, sensitive attribute names (i.e., users' demographic information) are hidden via one-hot encoding [16].

for changing its value in turn based on G_{local} (Line4-11). In this process, FairAS samples T distinct values from the value set for the sensitive attribute s_i , and replaces the original value of s_i in G_{local} with each of the sampled T values, respectively (Line6-8). Note that if the size of a value set is smaller than the default number of exploration times for each attribute (i.e., T), we set T to the size of the value set minus one (Line5). In this way, FairAS obtains a set of groups constructed through local search on G_{local} , denoted as $\mathbb{G}$. Finally, FairAS integrates $\mathbb{G}$ into $\mathbb{G}_{explored}$, which will be used in the next iteration of history-guided adaptive sampling (Line12).

4 Study Design

In the study, we aim to address the following research questions (RQs):

- **RQ1:** How does FairAS perform in identifying the most disadvantaged and suboptimal disadvantaged groups?
- **RQ2:** How does FairAS perform in improving the overall fairness of deep recommender systems with those identified disadvantaged groups?
- **RQ3:** Can each main component (i.e., preprocessing, local search enhancement, and history-guided adaptive sampling) contribute to the overall effectiveness of FairAS?

4.1 Subjects

In our study, we employed 24 subjects, encompassing four deep recommendation models trained on six datasets that are commonly used for recommendation tasks [14, 26, 41, 46]. Specifically, the 24 subjects resulted from training each deep learning recommendation model on the corresponding training data within each dataset. Due to the space limit, we uploaded the performance evaluation of these models, tested on the corresponding data, to our project homepage [2]. Our strategy to dividing the training and test sets follows the established methodology in the existing work [22].

The four recommendation models are Wide&Deep [12], DeepFM [20], FGCNN [28], and EDCN [8], hosted by either Google or Huawei. The six datasets to evaluate these models are *LFM360K* (a music recommendation dataset from Last.FM.) [7], *MovieLens* (a movie recommendation dataset from IMDB) [21], *BlackFriday* (a product recommendation dataset released by Kaggle for a contest) [1], *Amazon* (another product recommendation dataset from Amazon) [42], *Reasoner* (a micro-video recommendation dataset from Huawei) [10], and *Kuai* (another micro-video recommendation dataset from Kuaishou) [16]. Table 1 provides an overview of these datasets, including the number of users, the number of user groups, and the sensitive attributes involved during model training. These datasets exhibit remarkable diversity, spanning various fields like music, movies, micro-videos, and products, featuring different items and user scales, and a range of sensitive attributes.

4.2 Compared Approaches

In our study, we considered two compared approaches, i.e., TestSGD [49] and FairRec [19]. **TestSGD**, originally designed for deep learning classification models, controls the search process with a hyper-parameter θ , guiding to filter out the groups with fewer than θ percent of the total users. It then calculates the fairness scores for the remaining groups and identifies the most disadvantaged group through traversal. Following the existing work [19], we set θ to 0.005. **FairRec** is currently the state-of-the-art fairness testing approach tailored for deep recommender systems. As mentioned earlier, FairRec utilizes a global optimization algorithm, specifically DPSO, to identify the most disadvantaged group for a deep recommender system.

We also employed Themis [15], which adopts brute-force search to enumerate all user groups and obtain their corresponding values in terms of an evaluation metric. In this way, we can obtain the **ground-truth** results for the most disadvantaged group and all the sub-optimal disadvantaged groups. Although brute-force search is very costly, it is quite helpful to measure the gap between the effectiveness achieved by a fairness testing approach and the ideal effectiveness.

FairAS contains three key components: data preprocessing, local search enhancement, and history-guided adaptive sampling. We conducted an ablation study to investigate the contribution of each of them. Specifically, we constructed three variants of FairAS: **FairAS_{pre}** (removing data preprocessing from FairAS), **FairAS_{local}** (removing local search enhancement from FairAS), and **FairAS_{history}** (replacing history-guided adaptive sampling with uniform sampling in FairAS).

4.3 Implementations

We implemented FairAS in Python based on Pandas 1.3.2 [3]. We set the threshold τ for filtering out invalid groups to 0.1% of the user population in the component of data preprocessing following the existing work [19]. We set the number of groups sampled in each iteration (i.e., k) to 10 in the component of adaptive sampling. We set the number of exploration times for each attribute (i.e., T) to 5 in the component of local search enhancement. Regarding the baselines, we directly used their released implementations and recommended configurations in our study. To reduce the influence of randomness in our study, we repeated all the experiments five times and calculated the average results. Our experiments were conducted on the Intel Xeon Silver 4214 machine with 256GB RAM.

5 Results and Analysis

5.1 RQ1: Effectiveness of FairAS in Identifying the Most Disadvantaged and Sub-optimal Disadvantaged Groups

5.1.1 Setup. Following the existing work [19], we adopted the five evaluation metrics, (i.e., AUC, MRR, NDCG, URD, and URP) to measure the recommendation quality and evaluate disadvantaged groups, respectively. For each studied approach under each evaluation metric, we measured the metric value for each visited group, and denoted the group with the worst metric value as the most disadvantaged group identified by the fairness testing approach during the testing process. For ease of presentation, we denoted these values of the most disadvantaged group as M_{AUC} , M_{MRR} , M_{NDCG} , M_{URD} , M_{URP} , corresponding to AUC, MRR, NDCG, URD, and URP, respectively. A smaller value for M_{AUC} , M_{MRR} , M_{NDCG} , M_{URD} indicates better effectiveness of the fairness testing approach in identifying the most disadvantaged group, while a larger M_{URP} implies better effectiveness. Then, under each evaluation metric, we identified the groups with the Top- $n\%$ worst metric values among all the visited groups by a fairness testing approach as the suboptimal disadvantaged groups identified by this approach, and set $n\%$ to 10% following the existing work [19]. Additionally, we recorded the execution time required for a fairness testing approach to reach its terminating

Table 2. Comparison among Themis, TestSGD, FairRec, and FairAS in identifying the most disadvantaged groups under different evaluation metrics (MovieLens, Amazon, BlackFriday and LFM360K).

		MovieLens					LFM360K				
		$M_{AUC} \downarrow$	$M_{MRR} \downarrow$	$M_{NDCG} \downarrow$	$M_{URD} \downarrow$	$M_{URP} \uparrow$	$M_{AUC} \downarrow$	$M_{MRR} \downarrow$	$M_{NDCG} \downarrow$	$M_{URD} \downarrow$	$M_{URP} \uparrow$
Wide&Deep	Themis	0.7017	0.1041	0.0479	0.1091	0.0197	0.9110	0.6119	0.2243	0	0.0107
	TestSGD	0.8214	0.2298	0.1007	0.1545	0.0055	0.9243	0.6317	0.2303	0	0.0098
	FairRec	0.7506	0.1041	0.0479	0.1176	0.0095	0.9192	0.6126	0.2257	0	0.0098
	FairAS	0.7017	0.1041	0.0479	0.1091	0.0197	0.9110	0.6119	0.2243	0	0.0107
DeepFM	Themis	0.7207	0.0916	0.0539	0.1254	0.0218	0.9142	0.6102	0.2241	0	0.0108
	TestSGD	0.8357	0.2155	0.0910	0.1458	0.0023	0.9251	0.6318	0.2306	0	0.0099
	FairRec	0.7207	0.1392	0.0565	0.1357	0.0195	0.9178	0.6102	0.2241	0	0.0099
	FairAS	0.7207	0.0916	0.0539	0.1254	0.0218	0.9142	0.6102	0.2241	0	0.0108
FGCNN	Themis	0.6997	0.0861	0.0426	0.1272	0.0229	0.9050	0.6121	0.2232	0	0.0108
	TestSGD	0.8274	0.2057	0.0885	0.1407	0.0043	0.9241	0.6452	0.2316	0	0.0101
	FairRec	0.6997	0.1562	0.0699	0.1272	0.0150	0.9092	0.6243	0.2272	0	0.0105
	FairAS	0.6997	0.0861	0.0428	0.1272	0.0152	0.9050	0.6121	0.2232	0	0.0108
EDCN	Themis	0.5366	0.0357	0.0208	0.1272	0.0359	0.9098	0.5931	0.2184	0	0.0114
	TestSGD	0.5895	0.1108	0.0376	0.1605	0.0153	0.9206	0.6151	0.2251	0	0.0109
	FairRec	0.5591	0.0769	0.0329	0.1272	0.0308	0.9099	0.5967	0.2184	0	0.0114
	FairAS	0.5366	0.0416	0.0208	0.1272	0.0359	0.9099	0.5931	0.2184	0	0.0114
		BlackFriday					Amazon				
		$M_{AUC} \downarrow$	$M_{MRR} \downarrow$	$M_{NDCG} \downarrow$	$M_{URD} \downarrow$	$M_{URP} \uparrow$	$M_{AUC} \downarrow$	$M_{MRR} \downarrow$	$M_{NDCG} \downarrow$	$M_{URD} \downarrow$	$M_{URP} \uparrow$
Wide&Deep	Themis	0.7374	0.0416	0.0243	0.1166	0.0380	0.7523	0.2631	0.1033	0.0040	0.0101
	TestSGD	0.8946	0.3652	0.1518	0.2303	0.0048	0.7724	0.2855	0.1117	0.0073	0.0059
	FairRec	0.7511	0.1111	0.0814	0.1333	0.0287	0.7564	0.2631	0.1117	0.0055	0.0059
	FairAS	0.7384	0.0416	0.0243	0.1166	0.0380	0.7564	0.2631	0.1033	0.0055	0.0101
DeepFM	Themis	0.7063	0.0416	0.0243	0.0857	0.0354	0.7638	0.2772	0.1078	0.0055	0.0101
	TestSGD	0.9082	0.3533	0.1541	0.2027	0.0091	0.7782	0.2997	0.1174	0.0068	0.0061
	FairRec	0.7664	0.0571	0.0374	0.1285	0.0350	0.7673	0.2832	0.1078	0.0055	0.0061
	FairAS	0.7063	0.0416	0.0243	0.0857	0.0350	0.7638	0.2772	0.1078	0.0063	0.0101
FGCNN	Themis	0.7444	0.0416	0.0243	0.0857	0.0479	0.7565	0.2801	0.1098	0.0044	0.0102
	TestSGD	0.9121	0.3532	0.1435	0.2166	0.0048	0.7681	0.2988	0.1163	0.0075	0.0060
	FairRec	0.7765	0.0416	0.0243	0.0888	0.0314	0.7586	0.2801	0.1114	0.0044	0.0060
	FairAS	0.7444	0.0416	0.0243	0.0888	0.0479	0.7575	0.2801	0.1098	0.0044	0.0102
EDCN	Themis	0.6434	0.0333	0.0218	0.1142	0.0442	0.7548	0.2761	0.1087	0.0049	0.0102
	TestSGD	0.8457	0.3407	0.1360	0.2363	0.0089	0.7717	0.2838	0.1116	0.0075	0.0061
	FairRec	0.6621	0.0416	0.0243	0.1166	0.0409	0.7554	0.2820	0.1116	0.0055	0.0102
	FairAS	0.6434	0.0333	0.0218	0.1142	0.0409	0.7548	0.2761	0.1087	0.0055	0.0102

condition (i.e., the identified most disadvantaged group do not change for five consecutive iterations), providing a measure of its efficiency.

5.1.2 Results. Effectiveness of Identifying the Most Disadvantaged Group: Table 2 and 3 show the effectiveness of Themis, TestSGD, FairRec, and FairAS in identifying the most disadvantaged group for each studied deep recommender system under different evaluation metrics. We marked the best result among TestSGD, FairRec, and FairAS on each subject under each evaluation metric as bold in this table. From Table 2 and 3, among 120 cases (4 models $\times$ 6 datasets $\times$ 5 evaluation metrics), FairAS performs better than FairRec in 62.5% (75 out of 120) cases, performs equally to FairRec in 35.8% (43 out of 120) cases, and performs worse than FairRec in only 1.7% (2 out of 120) cases. Particularly, both FairAS and FairRec largely outperform TestSGD in almost all 120 cases. On average, FairAS improves by 19.2% over FairRec and by 37.77% over TestSGD across all 120 cases.

Table 3. Comparison among Themis, TestSGD, FairRec, and FairAS in identifying the most disadvantaged groups under different evaluation metrics (Reasoner and Kuai).

		Reasoner					Kuai				
		$M_{AUC} \downarrow$	$M_{MRR} \downarrow$	$M_{NDCG} \downarrow$	$M_{URD} \downarrow$	$M_{URP} \uparrow$	$M_{AUC} \downarrow$	$M_{MRR} \downarrow$	$M_{NDCG} \downarrow$	$M_{URD} \downarrow$	$M_{URP} \uparrow$
Wide&Deep	Themis	0.7199	0	0.0162	0.4666	0.0056	0.8935	0.8750	0.2967	0	0.0122
	TestSGD	0.8945	0.15	0.0549	0.7125	0.0043	0.9992	1.0	0.3572	0	0.0062
	FairRec	0.7607	0	0.0162	0.5467	0.0056	0.9340	0.9285	0.3391	0	0.0077
	FairAS	0.7199	0	0.0162	0.5333	0.0056	0.9063	0.8862	0.3162	0	0.0092
DeepFM	Themis	0.7494	0	0.0145	0.4666	0.0066	0.9232	0.8334	0.2875	0	0.0121
	TestSGD	0.8936	0.1333	0.0452	0.7250	0.0039	0.9997	1.0	0.3391	0	0.0075
	FairRec	0.7721	0.0572	0.0145	0.4800	0.0066	0.9410	0.8750	0.3014	0	0.0085
	FairAS	0.7494	0.0400	0.0145	0.4666	0.0066	0.9270	0.8639	0.2967	0	0.0088
FGCNN	Themis	0.7630	0	0.0188	0.4666	0.0071	0.9119	0.8750	0.2967	0	0.0099
	TestSGD	0.8667	0.1666	0.0594	0.6736	0.0053	0.9995	0.9817	0.3344	0	0.0071
	FairRec	0.7630	0.0400	0.0188	0.5066	0.0071	0.9711	0.8846	0.2967	0	0.0073
	FairAS	0.7630	0	0.0188	0.5000	0.0071	0.9399	0.8819	0.3165	0	0.0093
EDCN	Themis	0.6521	0	0.0028	0.4666	0.0061	0.6945	0.2917	0.0887	0	0.0092
	TestSGD	0.8363	0.1333	0.0452	0.7764	0.0043	0.8711	0.4256	0.1439	0	0.0050
	FairRec	0.6521	0	0.0032	0.4666	0.0052	0.8146	0.5026	0.1011	0	0.0061
	FairAS	0.6521	0	0.0028	0.4666	0.0054	0.7328	0.3950	0.0991	0	0.0073

Besides, the effectiveness of FairAS is equal to the ideal effectiveness (achieved by Themis) in 73.3% cases, while FairRec achieves this in only 31.7% cases. The results demonstrate the superiority of FairAS over FairRec in precisely identifying the most disadvantage group for deep recommender systems, even though FairRec directly sets the most disadvantaged group as its objective based on global optimization. Furthermore, we conducted a paired sample Wilcoxon signed-rank test [44] at a significance level of 0.05 to assess whether FairAS significantly outperforms FairRec across all datasets. We then applied the Vargha-Delaney effect size measure [39] to investigate the difference degree between FairAS and FairRec. Following the existing work [30], the difference degree is characterized as small, medium, and large when the effect size is larger than 0.56, 0.64, and 0.71, respectively. The results show that the p-values for five (out of six) datasets are smaller than 0.05, indicating the statistically significant superiority of FairAS over FairRec. The effect size values range from 0.56 to 0.71, because both FairRec and FairAS successfully identify the most disadvantaged group in some cases.

Based on the terminating condition, Table 4 shows the average time spent by Themis, TestSGD, FairRec, and FairAS on each subject across all five metrics, respectively. From Table 4, FairAS outperforms FairRec, TestSGD, and Themis in terms of efficiency, with average improvements of 43.07%, 53.70%, and 73.56% across all subjects, respectively. By referring to Table 2 and Table 4 together, we can conclude that FairAS almost always spends less time to achieve better or equal effectiveness than all compared approaches in identifying the most advantaged group. For example, on DeepFM with LFM360K under the AUC metric, FairAS spends 773 seconds to achieve the M_{AUC} value of 0.9142 (the same as the ideal effectiveness achieved by Themis, taking 8,733 seconds), while FairRec spends 1,798 seconds to achieve the M_{AUC} value of 0.9178. The results show high efficiency of FairAS. Furthermore, for the two cases where FairAS performs worse than FairRec in Table 2, when we ran FairAS for the same time as FairRec, FairAS can also perform equally to FairRec.

We further performed fine-grained analysis to demonstrate the superiority of FairAS over the best baseline (i.e., FairRec) in identifying the most disadvantaged group by taking the four subjects using the LFM360K dataset under AUC as the representative. The conclusions are consistent across different subjects and evaluation metrics, which can be found on our project homepage [2]. Figure 3

Table 4. Comparison among Themis, TestSGD, FairRec, and FairAS in terms of time efficiency on average across all five evaluation metrics (in seconds).

	MovieLens				LFM360K			
	Wide&Deep	DeepFM	FGCNN	EDCN	Wide&Deep	DeepFM	FGCNN	EDCN
Themis	33.22	31.39	37.46	31.48	8848.09	8733.56	8281.03	8687.49
TestSGD	18.99	20.98	29.06	20.55	1256.93	1197.87	1747.97	1288.01
FairRec	19.90	19.59	25.67	20.28	1766.17	1782.63	1997.68	1767.74
FairAS	18.70	18.72	23.77	20.23	706.58	564.04	909.77	978.61

	BlackFriday				Amazon			
	Wide&Deep	DeepFM	FGCNN	EDCN	Wide&Deep	DeepFM	FGCNN	EDCN
Themis	71.83	78.47	76.63	83.09	692.11	718.97	767.23	714.35
TestSGD	48.30	46.99	63.00	46.13	560.91	532.43	497.67	492.89
FairRec	34.93	45.04	31.57	29.81	554.24	526.27	457.16	390.51
FairAS	28.21	25.95	30.15	27.08	361.06	341.44	405.16	360.11

	Reasoner				Kuai			
	Wide&Deep	DeepFM	FGCNN	EDCN	Wide&Deep	DeepFM	FGCNN	EDCN
Themis	2718.36	2597.76	2547.67	2173.19	439627.38	543140.27	497233.29	497277.84
TestSGD	1952.72	1960.94	1999.17	1984.69	387003.70	384166.40	394826.01	396027.82
FairRec	146.12	165.16	161.62	112.31	2304.92	2413.38	2336.02	1901.74
FairAS	34.34	32.32	34.44	41.95	1572.34	1512.23	1693.52	1082.34

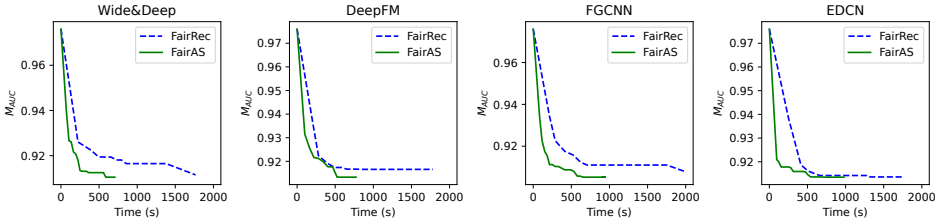


Fig. 3. The most disadvantaged group identified by FairAS and FairRec over time on LFM360K.

shows the effectiveness of FairAS and FairRec in terms of M_{AUC} during the overall running process of each approach. From Figure 3, FairAS achieves better effectiveness than FairRec at almost every time point. Further, we measured the time spent by FairAS on achieving the same UF_{AUC} value as the one achieved by FairRec, and found that FairAS saves 41.01%~71.06% running time compared to FairRec for the four subjects in Figure 3. The results further demonstrate the effectiveness and efficiency of FairAS in identifying the most disadvantaged group.

Effectiveness of Identifying Suboptimal Disadvantaged Groups: To measure the effectiveness of each approach in identifying suboptimal disadvantaged groups, we measured how close the suboptimal disadvantaged groups identified by the approach is to the ground-truth sub-optimal disadvantaged groups determined by Themis by calculating the differences between their metric values. Then, we drew box-plots to show the difference distribution for each approach. We used AUC as the representative metric and put the results in terms of the other four evaluation metrics on our project homepage [2]. Note that the conclusions are consistent across different evaluation metrics. Figure 4 illustrate the difference distributions for Top-10% suboptimal disadvantaged groups identified by FairAS and FairRec. The smaller the difference is (that is, the lower the box position is), the more accurate the identification of the sub-optimal disadvantaged groups is. We put the results of TestSGD at our project homepage [2] due to (1) the space limit and (2) both FairAS and FairRec always largely outperforming TestSGD.

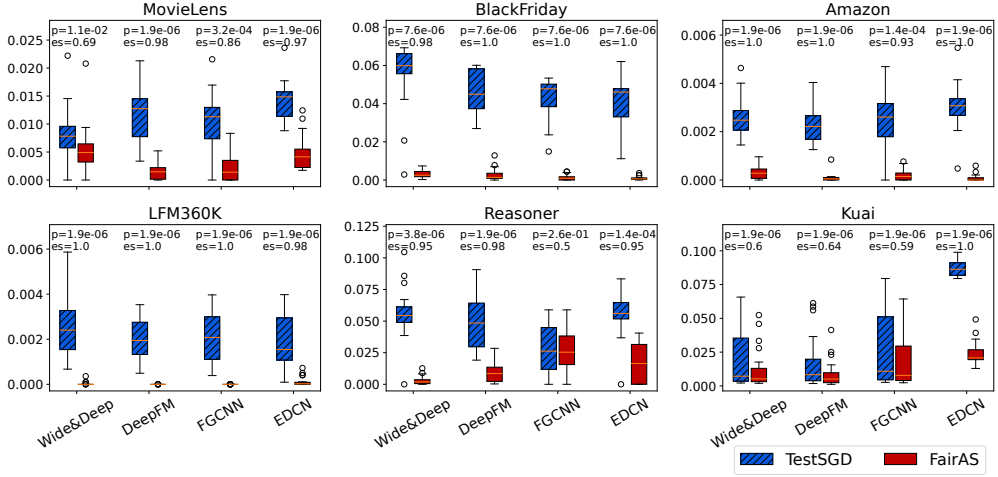


Fig. 4. Difference distributions for Top-10% suboptimal disadvantaged groups identified by FairAS and FairRec under the AUC metric.

From Figure 4, FairAS almost always has smaller differences than FairRec in terms of all of first quartile, median, and third quartile, demonstrating that FairAS can identify the sub-optimal disadvantaged groups more accurately. We further conducted a paired-sample Wilcoxon signed-rank test [44] at a significance level of 0.05 to investigate whether the superiority of FairAS over FairRec is statistically significant based on each pair of comparison in Figure 4, and also performed the Vargha-Delaney effect size measure [39] to assess the degree of the superiority accordingly. We showed the p-value (denoted as p in figures) and the effect-size value (denoted as es) on each pair of comparisons at the top of the corresponding boxes in Figure 4. We found that the p-values in 95.83% cases are smaller than 0.05, indicating that FairAS significantly outperforms FairRec in identifying suboptimal disadvantaged groups. The effect-size values in 79.17% cases are larger than 0.71 (this threshold follows the existing work [30]), indicating the large degree of the superiority of FairAS over FairRec.

Answer to RQ1: FairAS is more effective and efficient than the state-of-the-art FairRec in identifying both the most disadvantaged (with the average effectiveness improvement of 19.2% by saving 43.07% time) and suboptimal advantaged groups (with 79.17% cases achieving the statistically significant superiority with large degree).

5.2 RQ2: Effectiveness of FairAS in Improving System Fairness

5.2.1 Setup. Regarding the fairness improvement, we employed the widely-used post-processing approach [17] to improving the fairness of a deep recommender system based on those identified sub-optimal disadvantaged groups by FairRec and FairAS, following the existing work [19]. Here, we did not study TestSGD due to its significantly poor effectiveness in identifying disadvantaged groups compared to both FairRec and FairAS. This approach re-ranks the recommendation list for the identified suboptimal disadvantaged groups to optimize the recommendation effectiveness in terms of an evaluation metric. Then, we reported the improved fairness of deep recommender systems by each fairness testing approach for comparison.

Table 5. Comparison between FairAS and FairRec in fairness improvement under the MRR metric ($\uparrow$).

			MovieLens	Amazon	Reasoner	BlackFriday	LFM360K	Kuai
Wide&Deep	FairRec	Before	0.1786	0.2997	0.2181	0.2373	0.6419	0.9507
		After	0.3562	0.4560	0.3398	0.4567	0.6691	0.9724
	FairAS	Before	0.1720	0.2895	0.1284	0.1571	0.6408	0.9486
		After	0.4123	0.4681	0.4250	0.5572	0.7204	0.9962
DeepFM	FairRec	Before	0.1783	0.3023	0.1482	0.2185	0.6451	0.9587
		After	0.3847	0.4168	0.4173	0.4558	0.6555	0.9708
	FairAS	Before	0.1648	0.3012	0.1395	0.1319	0.6354	0.9182
		After	0.4131	0.4891	0.4447	0.4891	0.7054	0.9829
FGCNN	FairRec	Before	0.1998	0.2973	0.1477	0.2405	0.6461	0.9637
		After	0.3715	0.3902	0.3592	0.4827	0.6588	0.9683
	FairAS	Before	0.1732	0.2920	0.1407	0.1239	0.6431	0.9274
		After	0.3947	0.4056	0.4173	0.5266	0.6796	0.9757
EDCN	FairRec	Before	0.1596	0.2993	0	0.2438	0.6293	0.2516
		After	0.2597	0.4387	0.3000	0.4821	0.6419	0.4790
	FairAS	Before	0.1235	0.2938	0	0.1132	0.6252	0.1904
		After	0.3177	0.4563	0.3000	0.5408	0.6651	0.5428

5.2.2 Results. Table 5 presents the effectiveness of each subject in terms of each evaluation metric before and after applying the post-processing approach (denoted as Before and After in Table 5), each value in which represents the average value across the set of suboptimal disadvantaged groups identified by each approach. Due to the space limit, we used the MMR metric as the representative for illustration, and the conclusions across all the evaluation metrics are consistent. The complete results are available on our project homepage [2]. From Table 5, FairAS almost always achieves larger improvements than FairRec with the set of identified suboptimal disadvantaged groups. Note that the improvement refers to the difference between the evaluation metric value after applying the post-processing approach and that before applying the post-processing approach for a case in the table. On average, FairAS outperforms FairRec by 70.27% in terms of fairness improvement across all the cases. This is as expected, since the former is able to identify the suboptimal disadvantaged groups closer to the ground-truth ones, which tend to correspond to more severe fairness issues and thus have larger improvement room. Furthermore, the Wilcoxon signed-rank test at the significance level of 0.05 confirms that all p-values are smaller than 0.05, demonstrating the statistically significant superiority of FairAS over FairRec. All the Vargha-Delaney effect size values are larger than 0.64 (even 0.71 in MovieLens, BlackFriday, Amazon, and Kuai), demonstrating the medium or large degree of superiority.

Answer to RQ2: With those identified suboptimal disadvantaged groups, FairAS can more significantly improve recommender system fairness than FairRec across all subjects and evaluation metrics, with the average fairness improvement of 70.27%.

5.3 RQ3: Contributions of Key Components in FairAS

5.3.1 Setup. We compared FairAS and its three variants presented in Section 4.2 to investigate the contribution of each main component in FairAS (i.e., data preprocessing, local search enhancement, and history-guided sampling). Here, we compared them following the experimental setting in RQ1.

Table 6. Average effectiveness comparison among FairAS, FairAS_{!pre}, FairAS_{!local}, and FairAS_{!history} in identifying the most disadvantaged groups under the AUC metric ($\downarrow$) across the four subjects on each dataset.

	MovieLens	LFM360K	BlackFriday	Amazon	Reasoner	Kuai
FairAS	0.6646	0.9100	0.7081	0.7581	0.7211	0.8765
FairAS _{!pre}	0.7004	0.9125	0.7230	0.7597	0.7620	0.9151
FairAS _{!local}	0.6970	0.9115	0.7249	0.7594	0.7348	0.8880
FairAS _{!history}	0.7029	0.9121	0.7242	0.7597	0.7352	0.8881

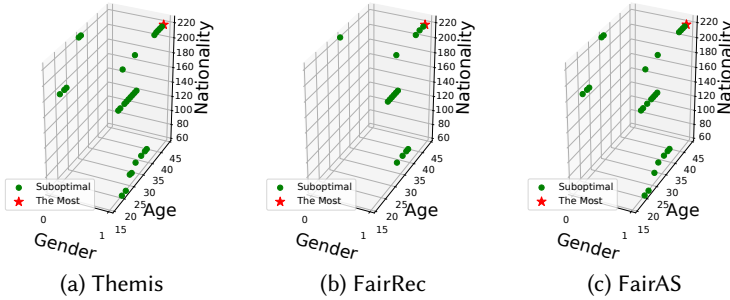


Fig. 5. Examples of disadvantaged groups identified by Themis, FairRec and FairAS on the Wide&Deep model trained with the LFM360K dataset.

5.3.2 Results. Table 6 shows the average results of FairAS and its three variants in identifying the most disadvantaged group across the four subjects on each dataset, where the best results are marked as bold among the four studied approaches. Due to the space limit, we used the AUC metric as the representative, and put the other metrics' results on the project homepage [2], which share consistent conclusions. From Table 6, FairAS performs the best in identifying the most disadvantaged group on all the subjects, with the average improvements of 28.8%, 17.3%, and 18.7% over FairAS_{!pre}, FairAS_{!local}, and FairAS_{!history}, respectively. Additionally, we investigated the effectiveness of FairAS and its three variants in identifying Top-10% suboptimal disadvantaged groups. The detailed results are put at our project homepage [2] due to the space limit. We found that the difference between the suboptimal disadvantaged groups identified by FairAS and the ground-truth ones is the smallest among these studied approaches on each subject, demonstrating the superiority of FairAS over its variants in identifying suboptimal disadvantaged groups. Overall, the experimental results demonstrate that all the three components contribute to the overall effectiveness of FairAS significantly. Besides, we found that FairAS_{!pre} performs much worse on more large-scale datasets (i.e., Reasoner and Kuai) from Table 6, demonstrating the significant contribution of data preprocessing on such subjects. This further confirms data preprocessing's ability to reduce the testing space from the entire space to that formed by valid groups. That is, as the subject grows in scale, FairAS benefits more from the component of data preprocessing.

Answer to RQ3: All the three main components (i.e., data preprocessing, local search enhancement, and history-guided sampling) contribute to the overall effectiveness of FairAS in identifying both the most disadvantaged and suboptimal disadvantaged groups. For more large-scale subjects, the contribution of data preprocessing becomes much more significant.

5.4 Case Study

We performed a case study for further illustration, especially qualitative analysis. Specifically, Figure 5 shows the disadvantaged groups identified by Themis, FairRec, and FairAS, on the Wide&Deep model trained with the LFM360K dataset. In this figure, the red star marks the identified most disadvantaged group, and the green points mark the identified suboptimal disadvantaged groups. Through observations from this figure, the challenges in identifying disadvantaged groups are further highlighted. Particularly, the result of Themis indicates that the suboptimal disadvantaged groups are not entirely located around the most disadvantaged group. Some suboptimal disadvantaged groups are located far from the most disadvantaged group. This also indicates the limitation of the global-optimization-based approaches (such as the state-of-the-art FairRec) in identifying suboptimal disadvantaged groups.

Let us use the example for further illustrating the challenge. In Figure 5a, the point (1, 42, 225) represents the most disadvantaged group, in which all users share the same attribute values (gender=1, age=42, nationality=225), corresponding to 42-year-old Mexican females. This may result from the limited interaction between users in this group and the system, which leads to insufficient training data and suboptimal recommendation quality for this specific group. In suboptimal disadvantaged groups, there are instances like (1, 40, 225) and (1, 38, 225), which are closer to the most disadvantaged group in terms of Euclidean distance within the space. There are also instances like (1, 21, 56) and (0, 22, 176), which are farther from the most disadvantaged group in terms of Euclidean distance. Therefore, when the optimization objective is just to identify the most disadvantaged group (like FairRec), it becomes difficult to find suboptimal disadvantaged groups that are far from the most disadvantaged group, indicating the challenge in this task.

We also performed overlap analysis on the suboptimal disadvantaged groups identified by FairAS and FairRec across all datasets. There are 23.95% suboptimal disadvantaged groups that are identified by both approaches. Besides, 65.70% suboptimal disadvantaged groups are only identified by FairAS, but just 7.08% suboptimal disadvantaged groups are only identified by FairRec. Indeed, from Figure 5, most of the suboptimal disadvantaged groups overlooked by FairRec are located far from the most disadvantaged group. However, FairAS, based on adaptive sampling, is able to effectively identify such suboptimal disadvantaged groups. This demonstrates the superiority of FairAS over FairRec in identifying suboptimal disadvantaged groups to a large extent.

5.5 Threats to Validity

The threat to *internal* validity mainly lies in the implementations of FairAS and the compared approaches. To reduce it, we adapted the implementations of the compared approaches released by the authors, implemented FairAS based on some existing libraries (presented in Section 4.3), and carefully checked the code of our approach FairAS and experimental scripts.

The threat to *external* validity mainly lies in used subjects. To reduce it, we used all the subjects that have been used in the study of evaluating FairRec [19] and additionally used two more large-scale datasets (i.e., Reasoner and Kuai) to further increase the diversity of subjects.

The threats to *construct* validity mainly lie in randomness, hyper-parameter settings in FairAS, and the step of filtering out groups with few users. To reduce the threat from randomness, we have repeated all the experiments five times and calculated the average results.

To reduce the threat from hyper-parameter settings, we first conducted an experiment to investigate the influence of two key hyper-parameters (i.e., k and T) in FairAS, taking the four subjects on the LFM360K dataset with the AUC metric as the representative. Specifically, k (with a default value of 10) refers to the number of groups sampled in each iteration in the adaptive sampling component, and T (with a default value of 5) refers to the number of exploration times for each

Table 7. Effectiveness of FairAS with different hyper-parameter settings in identifying the most disadvantaged group on the LFM360K under the AUC metric ($\downarrow$).

	k					T				
	2	5	10	15	20	3	4	5	6	7
Wide&Deep	0.9187	0.9152	0.9110	0.9110	0.9110	0.9126	0.9110	0.9110	0.9110	0.9110
DeepFM	0.9178	0.9163	0.9142	0.9142	0.9142	0.9151	0.9142	0.9142	0.9142	0.9142
FGCNN	0.9083	0.9050	0.9050	0.9050	0.9050	0.9214	0.9050	0.9050	0.9050	0.9050
EDCN	0.9103	0.9099	0.9099	0.9099	0.9098	0.9099	0.9099	0.9099	0.9099	0.9099

Table 8. Time efficiency of FairAS with different hyper-parameter settings on the LFM360K (in seconds).

	k					T				
	2	5	10	15	20	3	4	5	6	7
Wide&Deep	435.65	648.55	706.58	722.30	927.76	692.09	706.15	706.58	709.65	712.79
DeepFM	419.91	586.94	564.04	707.35	919.27	560.06	560.30	564.04	564.98	570.80
FGCNN	455.64	698.47	909.77	844.75	909.82	901.42	905.55	909.77	914.63	920.66
EDCN	506.34	768.59	978.61	993.56	981.24	951.74	970.29	978.61	980.22	980.45

attribute in the local search enhancement component. We studied different k values (i.e., 2, 5, 10, 15, 20) and T values (i.e., 3, 4, 5, 6, 7) in this experiment, respectively. Table 7 shows the effectiveness of FairAS under different settings in identifying the most disadvantaged group, and Table 8 shows the time spent by FairAS under different settings. The results show that as the value of k increases, the identification of the most disadvantaged group becomes more accurate. However, this also leads to more time cost. When setting k to 10 (our default setting), the effectiveness of FairAS is very close to that under $k = 20$, but its time cost is much less than the latter. Therefore, we balance the effectiveness and efficiency to set the default value for k . Regarding T , except $T = 3$, the effectiveness and time cost of FairAS under the remaining settings are very close, indicating that FairAS is stable with regard to T to a large degree. Additionally, we investigated the effectiveness of FairAS in identifying suboptimal disadvantaged groups when $n\%=20\%$, in order to further improve the evaluation generalizability. The conclusions are consistent, demonstrating the more precise identification with FairAS on suboptimal disadvantaged groups regardless of Top-10% or Top-20%. The detailed results are put at our project homepage [2] due to the space limit.

Regarding the last threat, the decision to exclude extremely small groups stems from the difficulty in determining whether a group is advantageous or disadvantageous. When the number of users in a group is very small, it becomes challenging to draw statistically reliable conclusions about the potential biases within that group [36]. To assess whether a group is disadvantaged, we compare its average recommendation quality to the overall metric for the entire population. The average value is calculated based on the accessible samples from that group, and only when the group size is sufficiently large can we ensure that the average estimation is accurate. If the sample size is too small, it may result in unreliable estimation of outcomes. Therefore, when a group has a very limited number of users, the calculated indicator of recommendation quality may not accurately represent the group's true recommendation quality. To clarify, these small groups emerge as the entire population is extensively divided based on various sensitive user attributes and the different values of each attribute. The process of refining this abstraction increases sensitivity to bias but also leads to the formation of very small groups. This presents a dilemma encountered in related works; for instance, Guo et al. [19] classified groups with fewer than 0.1% of the total population as invalid and filtered them out. We adopt this method in our work. As a result, the evaluation

outcomes are expected to be more reliable, as we focus only on groups that clearly exhibit either advantages or disadvantages.

6 Related Work

Fairness Testing of Recommender Systems. Our work belongs to fairness testing of recommender systems. The most related work is FairRec [19], which is the state-of-the-art approach specific to deep recommender systems and has been discussed and empirically compared in Sections 4.2 and 5. Specifically, FairAS and FairRec differ primarily in their search goals and methodologies. FairRec focuses on identifying the most disadvantaged group using DPSO, a global optimization method that iteratively updates particles (user groups) toward the global optimum. However, it overlooks suboptimal groups and suffers from slow convergence. In contrast, FairAS seeks both the most disadvantaged and suboptimal disadvantaged groups, employing an adaptive sampling strategy that adjusts probabilities based on past discoveries and uses local search to accelerate the identification process. Besides, Mehrotra et al. [31] conducted a fairness assessment by quantifying differences in user satisfaction among various single-attribute demographic groups. Li et al. [25] categorized users based on their activity levels and discovered that inactive users are more susceptible to unfair treatment. Wu et al. [45] introduced a sensitive attribute predictor to assess the relationship between news recommendations and gender, employing it to assess the fairness of news recommender systems. However, these existing approaches focused on evaluating fairness on single sensitive attribute rather than multiple attributes (which is the significant characteristic of deep recommend systems as presented in the existing work [19]). Different from them, FairAS targets this more challenging problem, i.e., fairness testing of deep recommender systems with multiple attributes (same as FairRec).

Fairness Testing of Other Types of Deep Learning Systems. Besides fairness testing of deep recommender systems, there are also some approaches proposed for testing the fairness of other kinds of deep learning models (such as classification models) [5, 11, 15, 34, 38, 49, 50]. To assess individual fairness, some studies [32, 50, 52] generate pairs of discriminatory instances that differ only in sensitive attributes to identify fairness issues within classification models. For example, Udeshi et al. [38] proposed AEQUITAS, a method that combines global and local search to identify discriminatory instances. Specifically, the global search phase employs random search. Once a discriminatory instance is found, the attribute value is perturbed to explore more discriminatory cases in its vicinity. AEQUITAS dynamically adjusts the selection probabilities for two attribute value changing rules, and selects a specific rule to change the target attribute value for local search. Different from AEQUITAS, FairAS performs adaptive sampling in the global search phase, which dynamically adjusts the sampling probabilities of all attribute values based on exploration history. Such an adaptive sampling strategy in the global search phase can more efficiently explore the vast search space, aligning with the goal of group fairness testing. Zhang et al. [50] introduced ADF that performs global search and local search based on gradient-based guidance. However, it is applicable to calculate gradients for each individual input, but is inapplicable to group fairness testing [50].

In addition to research focusing on individual fairness, there exist studies targeting group fairness [5, 15, 49]. For instance, Si et al. [34] proposed a statistical testing framework that evaluates the fairness of classification models through optimal transport and linear programming. Different from them, our work concentrates on testing the fairness of deep recommender systems rather than general classification models, which involve more vast and sparse testing space. Additionally, fairness metrics for deep recommender systems differ fundamentally from general classification models. In particular, the existing work has demonstrated that these approaches proposed for classification models cannot perform well in fairness testing of deep recommender systems [19].

Search-based Fairness Improvement. There are also some studies that employ search algorithms in the area of fairness improvement [13, 37, 47]. We also discuss the differences between them and our work in designing search algorithms. In general, the search algorithms used in them are global optimization methods with a single objective, such as the simulated annealing algorithm used by NeuFair [13] and the Pareto optimal search algorithm used by PairFair-ML [37]. They define a single optimization objective, such as identifying the most disadvantaged group in our task, and iteratively search for a global optimum. Therefore, they are very likely to overlook suboptimal disadvantaged groups in our task. Moreover, they suffer from a common drawback of global optimization methods, i.e., slow convergence [24], especially for the vast search space. Different from them, FairAS employs an adaptive sampling strategy that not only seeks the most disadvantaged group but also allocates probabilities to explore beyond the global optimum. Specifically, FairAS uses past discoveries as cues to uncover additional disadvantaged groups and leverages the clustering tendency of disadvantaged groups to enhance efficiency through local search.

7 Conclusion

In this paper, we presented FairAS, an innovative approach for fairness testing in deep recommender systems that effectively identifies both the most disadvantaged and suboptimal disadvantaged groups. FairAS leverages adaptive sampling, dynamically adjusting sampling probabilities based on exploration history, to guide the sampling process. By integrating local search with adaptive sampling, FairAS accelerates the discovery of clusters of disadvantaged groups, enabling more comprehensive fairness testing within a limited time. Our extensive evaluation on 24 configurations of deep recommendation models and datasets shows that FairAS outperforms the state-of-the-art approach, FairRec, with an average improvement of 19.2% in identifying the most disadvantaged groups and a 43.07% reduction in testing time. Additionally, FairAS enhances system fairness by 70.27% compared to FairRec on average across all cases.

8 Data Availability

We have released the implementation of FairAS and experimental data on our project homepage [2].

9 Ethical Considerations

This research investigates fairness testing in recommender systems, with the goal of identifying potential biases in algorithms. From an ethical standpoint, detecting unfairness in these systems is essential, as early identification of fairness issues can encourage developers to implement corrective measures, such as adjusting recommendation strategies. This proactive approach can help prevent the worsening of biases. Additionally, a fair recommender system offers users access to a wider range of information, which can help disrupt information bubbles and reduce societal polarization. To avoid ethical issues in our study, we adhered to established guidelines, utilized open-source datasets, ensured user anonymity through ID-based anonymization, and ensured that no human participants were directly involved.

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References

- [1] Accessed: 2023. BlackFriday. <https://www.kaggle.com/datasets/sdolezel/black-friday>.
- [2] Accessed: 2023. Homepage. <https://github.com/anonyProjects/FairAS>.
- [3] Accessed: 2023. Pandas. <https://pandas.pydata.org/>.
- [4] Himan Abdollahpour, Masoud Mansoury, Robin Burke, Bamshad Mobasher, and Edward Malthouse. 2021. User-centered evaluation of popularity bias in recommender systems. In *Proceedings of the 29th ACM Conference on User Modeling, Adaptation and Personalization*. 119–129.
- [5] Rico Angell, Brittany Johnson, Yuriy Brun, and Alexandra Meliou. 2018. Themis: Automatically testing software for discrimination. In *Proceedings of the 2018 26th ACM Joint meeting on european software engineering conference and symposium on the foundations of software engineering*. 871–875.
- [6] Eduard Baranov, Axel Legay, and Kuldeep S Meel. 2020. Baital: an adaptive weighted sampling approach for improved t-wise coverage. In *Proceedings of the 28th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering*. 1114–1126.
- [7] Òscar Celma Herrada et al. 2009. *Music recommendation and discovery in the long tail*. Universitat Pompeu Fabra.
- [8] Bo Chen, Yichao Wang, Zhirong Liu, Ruiming Tang, Wei Guo, Hongkun Zheng, Weiwei Yao, Muyu Zhang, and Xiuqiang He. 2021. Enhancing explicit and implicit feature interactions via information sharing for parallel deep ctr models. In *Proceedings of the 30th ACM international conference on information & knowledge management*. 3757–3766.
- [9] Junjie Chen, Zhuo Wu, Zan Wang, Hanmo You, Lingming Zhang, and Ming Yan. 2020. Practical accuracy estimation for efficient deep neural network testing. *ACM Transactions on Software Engineering and Methodology (TOSEM)* 29, 4 (2020), 1–35.
- [10] Xu Chen, Jingsen Zhang, Lei Wang, Quanyu Dai, Zhenhua Dong, Ruiming Tang, Rui Zhang, Li Chen, and Ji-Rong Wen. 2023. REASONER: An Explainable Recommendation Dataset with Multi-aspect Real User Labeled Ground Truths Towards more Measurable Explainable Recommendation. *CoRR* abs/2303.00168 (2023).
- [11] Zhenpeng Chen, Jie M Zhang, Max Hort, Mark Harman, and Federica Sarro. 2023. Fairness Testing: A Comprehensive Survey and Analysis of Trends. *ACM Transactions on Software Engineering and Methodology* (2023).
- [12] Heng-Tze Cheng, Levent Koc, Jeremiah Harmsen, Tal Shaked, Tushar Chandra, Hrishu Aradhye, Glen Anderson, Greg Corrado, Wei Chai, Mustafa Ispir, et al. 2016. Wide & deep learning for recommender systems. In *Proceedings of the 1st workshop on deep learning for recommender systems*. 7–10.
- [13] Vishnu Asutosh Dasu, Ashish Kumar, Saeid Tizpaz-Niari, and Gang Tan. 2024. NeuFair: Neural Network Fairness Repair with Dropout. In *Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis*. 1541–1553.
- [14] Michael D Ekstrand, Mucun Tian, Ion Madrazo Azpiazu, Jennifer D Ekstrand, Oghenemaro Anuyah, David McNeill, and Maria Soledad Pera. 2018. All the cool kids, how do they fit in?: Popularity and demographic biases in recommender evaluation and effectiveness. In *Conference on fairness, accountability and transparency*. PMLR, 172–186.
- [15] Sainyam Galthotra, Yuriy Brun, and Alexandra Meliou. 2017. Fairness testing: testing software for discrimination. In *Proceedings of the 2017 11th Joint meeting on foundations of software engineering*. 498–510.
- [16] Chongming Gao, Shijun Li, Wenqiang Lei, Jiawei Chen, Biao Li, Peng Jiang, Xiangnan He, Jiaxin Mao, and Tat-Seng Chua. 2022. KuaiRec: A Fully-Observed Dataset and Insights for Evaluating Recommender Systems. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management (Atlanta, GA, USA) (CIKM '22)*. 540–550. <https://doi.org/10.1145/3511808.3557220>
- [17] Elizabeth Gómez, Carlos Shui Zhang, Ludovico Boratto, Maria Salamó, and Mirko Marras. 2021. The winner takes it all: geographic imbalance and provider (un) fairness in educational recommender systems. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 1808–1812.
- [18] Antonio Guerriero, Roberto Pietrantuono, and Stefano Russo. 2021. Operation is the hardest teacher: estimating DNN accuracy looking for mispredictions. In *2021 IEEE/ACM 43rd International Conference on Software Engineering (ICSE)*. IEEE, 348–358.
- [19] Huizhong Guo, Jinfeng Li, Jingyi Wang, Xiangyu Liu, Dongxia Wang, Zehong Hu, Rong Zhang, and Hui Xue. 2023. FairRec: Fairness Testing for Deep Recommender Systems. In *ISSTA*. ACM, 310–321.
- [20] Huifeng Guo, Ruiming Tang, Yunming Ye, Zhenguo Li, and Xiuqiang He. 2017. DeepFM: a factorization-machine based neural network for CTR prediction. *arXiv preprint arXiv:1703.04247* (2017).
- [21] F Maxwell Harper and Joseph A Konstan. 2015. The movielens datasets: History and context. *Acm transactions on interactive intelligent systems (tiis)* 5, 4 (2015), 1–19.
- [22] Xiangnan He, Lizi Liao, Hanwang Zhang, Liqiang Nie, Xia Hu, and Tat-Seng Chua. 2017. Neural collaborative filtering. In *Proceedings of the 26th international conference on world wide web*. 173–182.
- [23] Kalervo Järvelin and Jaana Kekäläinen. 2002. Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems (TOIS)* 20, 4 (2002), 422–446.

- [24] Shutao Li, Minghui Tan, Ivor W Tsang, and James Tin-Yau Kwok. 2011. A hybrid PSO-BFGS strategy for global optimization of multimodal functions. *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)* 41, 4 (2011), 1003–1014.
- [25] Yunqi Li, Hanxiong Chen, Zuohui Fu, Yingqiang Ge, and Yongfeng Zhang. 2021. User-oriented Fairness in Recommendation. In *WWW. ACM / IW3C2*, 624–632.
- [26] Yunqi Li, Hanxiong Chen, Shuyuan Xu, Yingqiang Ge, and Yongfeng Zhang. 2021. Towards personalized fairness based on causal notion. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 1054–1063.
- [27] Charles X Ling, Jin Huang, Harry Zhang, et al. 2003. AUC: a statistically consistent and more discriminating measure than accuracy. In *Ijcai*, Vol. 3. 519–524.
- [28] Bin Liu, Ruiming Tang, Yingzhi Chen, Jinkai Yu, Huifeng Guo, and Yuzhou Zhang. 2019. Feature Generation by Convolutional Neural Network for Click-Through Rate Prediction. In *WWW. ACM*, 1119–1129.
- [29] Chuan Luo, Binqi Sun, Bo Qiao, Junjie Chen, Hongyu Zhang, Jinkun Lin, Qingwei Lin, and Dongmei Zhang. 2021. LS-Sampling: An Effective Local Search based Sampling Approach for Achieving High t-wise Coverage. In *Proceedings of ESEC/FSE 2021*. 1081–1092.
- [30] Ke Mao, Mark Harman, and Yue Jia. 2016. Sapienz: Multi-objective automated testing for android applications. In *Proceedings of the 25th international symposium on software testing and analysis*. 94–105.
- [31] Rishabh Mehrotra, Ashton Anderson, Fernando Diaz, Amit Sharma, Hanna Wallach, and Emine Yilmaz. 2017. Auditing search engines for differential satisfaction across demographics. In *Proceedings of the 26th international conference on World Wide Web companion*. 626–633.
- [32] Varya Monjezi, Ashutosh Trivedi, Gang Tan, and Saeid Tizpaz-Niari. 2023. Information-theoretic testing and debugging of fairness defects in deep neural networks. In *2023 IEEE/ACM 45th International Conference on Software Engineering (ICSE)*. IEEE, 1571–1582.
- [33] Lijing Qin and Xiaoyan Zhu. 2013. Promoting diversity in recommendation by entropy regularizer. In *Twenty-Third International Joint Conference on Artificial Intelligence*. Citeseer.
- [34] Nian Si, Karthyek Murthy, Jose H. Blanchet, and Viet Anh Nguyen. 2021. Testing Group Fairness via Optimal Transport Projections. In *ICML (Proceedings of Machine Learning Research, Vol. 139)*. PMLR, 9649–9659.
- [35] Jasper Snoek, Hugo Larochelle, and Ryan P Adams. 2012. Practical bayesian optimization of machine learning algorithms. *Advances in neural information processing systems* 25 (2012).
- [36] Steven K Thompson. 2012. *Sampling, Third Edition*. John Wiley & Sons, Inc.
- [37] Saeid Tizpaz-Niari, Ashish Kumar, Gang Tan, and Ashutosh Trivedi. 2022. Fairness-aware configuration of machine learning libraries. In *Proceedings of the 44th International Conference on Software Engineering*. 909–920.
- [38] Sakshi Udeshi, Pryanshu Arora, and Sudipta Chattopadhyay. 2018. Automated directed fairness testing. In *Proceedings of the 33rd ACM/IEEE International Conference on Automated Software Engineering*. 98–108.
- [39] András Vargha and Harold D Delaney. 2000. A critique and improvement of the CL common language effect size statistics of McGraw and Wong. *Journal of Educational and Behavioral Statistics* 25, 2 (2000), 101–132.
- [40] Ellen M Voorhees. 2001. The TREC question answering track. *Natural Language Engineering* 7, 4 (2001), 361–378.
- [41] Mengting Wan, Jianmo Ni, Rishabh Misra, and Julian McAuley. 2020. Addressing marketing bias in product recommendations. In *Proceedings of the 13th international conference on web search and data mining*. 618–626.
- [42] Wenjie Wang, Fuli Feng, Xiangnan He, Xiang Wang, and Tat-Seng Chua. 2021. Deconfounded recommendation for alleviating bias amplification. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*. 1717–1725.
- [43] Zan Wang, Ming Yan, Junjie Chen, Shuang Liu, and Dongdi Zhang. 2020. Deep learning library testing via effective model generation. In *Proceedings of the 28th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering*. 788–799.
- [44] Frank Wilcoxon. 1992. Individual comparisons by ranking methods. In *Breakthroughs in Statistics: Methodology and Distribution*. Springer, 196–202.
- [45] Chuhan Wu, Fangzhao Wu, Xiting Wang, Yongfeng Huang, and Xing Xie. 2021. Fairness-aware News Recommendation with Decomposed Adversarial Learning. In *AAAI*. AAAI Press, 4462–4469.
- [46] Le Wu, Lei Chen, Pengyang Shao, Richang Hong, Xiting Wang, and Meng Wang. 2021. Learning fair representations for recommendation: A graph-based perspective. In *Proceedings of the Web Conference 2021*. 2198–2208.
- [47] Junjie Yang, Jiajun Jiang, Zeyu Sun, and Junjie Chen. 2024. A large-scale empirical study on improving the fairness of image classification models. In *Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis*. 210–222.
- [48] Hanmo You, Zan Wang, Junjie Chen, Shuang Liu, and Shuochuan Li. 2023. Regression Fuzzing for Deep Learning Systems. In *ICSE*. IEEE, 82–94.

- [49] Mengdi Zhang, Jun Sun, Jingyi Wang, and Bing Sun. 2022. TestSGD: Interpretable Testing of Neural Networks Against Subtle Group Discrimination. *ACM Transactions on Software Engineering and Methodology* (2022).
- [50] Peixin Zhang, Jingyi Wang, Jun Sun, Guoliang Dong, Xinyu Wang, Xingen Wang, Jin Song Dong, and Ting Dai. 2020. White-box fairness testing through adversarial sampling. In *Proceedings of the ACM/IEEE 42nd International Conference on Software Engineering*. 949–960.
- [51] Shuai Zhang, Lina Yao, Aixin Sun, and Yi Tay. 2019. Deep learning based recommender system: A survey and new perspectives. *ACM computing surveys (CSUR)* 52, 1 (2019), 1–38.
- [52] Haibin Zheng, Zhiqing Chen, Tianyu Du, Xuhong Zhang, Yao Cheng, Shouling Ji, Jingyi Wang, Yue Yu, and Jinyin Chen. 2022. Neuronfair: Interpretable white-box fairness testing through biased neuron identification. In *Proceedings of the 44th International Conference on Software Engineering*. 1519–1531.
- [53] Cai-Nicolas Ziegler, Sean M McNee, Joseph A Konstan, and Georg Lausen. 2005. Improving recommendation lists through topic diversification. In *Proceedings of the 14th international conference on World Wide Web*. 22–32.

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